

marginal utility of money is constant for all utility differences of interest.^{bb}

GRADED-PAIRS ESTIMATES

Table 2 contains the definitions of the variables used in all analyses in this study. Attributes are effects-coded rather than dummy variable coded.^{cc} Consequently, the omitted categories NOSE and NOLIM are the negative sum of the included symptoms, interacted with the log of duration, and activity categories.^{dd} The symptom coefficients represent the disutility of the interacted symptom relative to the mean effect. The daily activity restriction levels represent the additional disutility of experiencing these restrictions compared with experiencing no limitations (NOLIM). We report both nominal and normalized coefficients. The normalized coefficients rescale the health index from 0 to 1, with the INHOSP coefficient equal to zero (the least utility) and the NOLIM coefficient equal to one (the most utility). The normalized coefficients facilitate comparisons of marginal utility across the symptom and activity levels within this model.

Table 3 reports estimates for the graded-pair, ordered-probit panel model.^{ee} This model estimates and controls for the correlation among the subject's eight SP answers due to unobserved heterogeneity across subjects. The correlation coefficient, RHO, shows that there is significant correlation among the eight SP questions. Controlling for this correlation affects the significance of several variables.^{ff}

Symptom attribute levels are interacted with the log of duration to facilitate comparisons with the discrete-choice estimates.^{gg} There is no natural ordering of symptom disutility, so we have no specific expectations about relative magnitudes. Nevertheless, it is not surprising that the coefficients on NOSE and EYE indicate higher utilities than the other symptoms, and these differences are significant at the 1% level.^{hh} Overall, there is relatively little variation in the normalized symptom coefficients compared with the activity level coefficients, as the normalized symptom coefficients range from 0.3535 to 0.6096. However, five of the seven symptoms are significantly different than the overall mean symptom effect. In addition, PAIN and BREATH have significantly less disutility than FLUTTER, ACHE and SWELL.

Unlike the symptom variables, the activity restriction levels have a natural ordering with 'no limitations' (NOLIM) representing the best activity restriction level and 'in hospital' (INHOSP) representing the worst activity restriction level. All of the coefficients on the activity restriction levels are significantly different than the mean activity effect. The coefficients of SOMELIM and NOSOC are reversed in order but are statistically equivalent, indicating that subjects did not have strong preferences for not being able to participate in social and recreational activities and having physical limitations versus simply having physical limitations. The remaining coefficients decrease monotonically, as expected, indicating that greater activity restrictions result in significantly greater utility losses.

Allowing the marginal utility of money δ^i to vary with personal characteristics allows for differences in tastes to affect the relative utilities of health and money, even though the β_j parameters are constant across subjects. The utility of money is the inverse of the disutility of cost. Thus, a positive coefficient on a covariate means that an increase in the variable moves the disutility of costs closer to zero, resulting in higher WTP. Alternatively, a negative coefficient results in lower WTP estimates. Except for VIMPORTANT and EDUCATION, the utility of money covariates are significant, suggesting that different groups of subjects have different preferences for health state/money trade-offs. We did not have specific expectations of the sign of SCORE. However, the results show that subjects with higher quiz scores have lower WTP for improved health, suggesting that subjects who accurately processed the information in the article are willing to pay less to avoid the specific health effects discussed. On the other hand, older and symptomatic subjects have higher WTP, possibly because their current health state is poor and thus marginal improvements are more valuable or because they want to avoid the recurrence of an illness.

The scale variables measure the amount of variance in the error term, indicating and controlling for the amount of 'noise' in the subject's ratings. Because the scale factor is inversely related to the variance of the error term, as the scale factor increases, the variance decreases. The significance of the scale parameters indicates that subjects used the rating scale differently. Of the significant coefficients in the scale function, subjects with

Table 2. Variables used in analysis

Variables	Description
NOSE	Stuffy or runny nose and sore throat
EYE	Eye irritation
FLUTTER	Fluttering in chest and feeling light-headed
BREATH	Coughing, wheezing, shortness of breath
ACHE	Coughing or wheezing with fever, chills or aching all over
SWELL	Shortness of breath and swelling in ankles and feet
PAIN	Pain in chest or arm
LNDAYS	Log of the number of days of episode (1, 5, 10) plus one
NOLIM	Activity level 1: You can go to work, go to school, do housework, participate in social or recreational activities and have no physical limitations
SOMELIM	Activity level 2: You can go to work, go to school, do housework and participate in social or recreational activities, but you have some physical limitations (trouble bending, stooping or doing vigorous activities) because of this health condition
NOSOC	Activity level 3: You can go to work, go to school, do housework, but you have some physical limitations (trouble bending, stooping or doing vigorous activities) and cannot participate in social or recreational activities because of this health condition
ATHOME	Activity level 4: You cannot leave your house, go to work, go to school, do housework, participate in social or recreational activities, and you have some physical limitations (trouble bending, stooping or doing vigorous activities) because of this health condition, but you can care for yourself
NEEDHELP	Activity level 5: You cannot leave your house, go to work, go to school, do housework, participate in social or recreational activities and you need help caring for yourself (feeding, bathing, dressing, toilet) because of this health condition
INHOSP	Activity level 6: You are in hospital and need help caring for yourself (feeding, bathing, dressing, toilet)
SQTCOST	Square root of the cost levels per year (CAN\$10–CAN\$750)
SCORE	Quiz score (percent correct)
AGE	The midpoint of the age category
AGESQUARED	The square of AGE
EDUCATION	Number of years of education
SYMPTOMATIC	Dummy variable = 1 if subject has ever been diagnosed with any cardiovascular or respiratory conditions or other serious illnesses
VIMPORTANT	Number of health factors subjects indicated as being very important to a person's health (0–4)
HISCORE	= 1 if subject scored 75% or better on the quiz
HIAGE	= 1 if subject is 60 years old or older
NOPAIDLEAVE	= 1 if subject is working but does not have paid sick leave
FAILCHECKS	= 1 if subject provided questionable or inconsistent answers in parts of the survey
HIINFORMATION	= 1 if subject acquires health information a few hours per week or more
ASTHMA	= 1 if subject has experienced asthma in past year
BRONCHITIS	= 1 if subject has experienced chronic bronchitis/ emphysema in past year
LUNGINFECTION	= 1 if subject has experienced lung infection (pneumonia, acute bronchitis) in past year
HEARTDISEASE	= 1 if subject has experienced heart disease symptoms in past year
INCOME	Household income before taxes in 1996
PAINRATING	Rating of pain or discomfort on a scale of 0 (no discomfort) to 8 (severe discomfort), summed over all symptoms
FREQILL	Frequency of illness on a scale of 0 (never) to 4 (almost all or all the time), summed over all symptoms
FREQHOME	Frequency of staying home because of illness in past year on a scale of 0 (never) to 4 (almost all or all the time), summed over all symptoms
CH_IC	Alternative specific dummy for the first alternative in each choice set (Initial Condition)
CH_A	Alternative specific dummy for the middle alternative in each choice set (Alternative A)
RHO	Random effects correlation coefficient for subject's eight SP questions

high quiz scores (75% or higher) and no paid sick leave had less noisy responses. The first result is consistent with the intuition that subjects who have digested and understood the health symptom background materials may have better-defined preferences for health–money trade-offs. However, we would have expected subjects with no paid leave to have noisier ratings, everything else equal. The survey told all subjects to assume that their lost ‘sick’ time would be covered by paid

sick leave. For those subjects who do not have paid sick leave in reality, we expected to observe noise in their ratings because the survey assumption conflicts with their real situation. However, because the instructions were different from their actual situation, these subjects may have been more aware of the instructions and thus more consistent in their ratings. Alternatively, NOPAIDLEAVE could be capturing other characteristics that influence noise, such as differences in the

Table 3. Graded-pair ordered-probit panel model

Variable	Coefficient	Normalized	<i>t</i> ratio
Health attributes			
NOSE*LNDAYS	0.2720***	0.6096	4.38
EYE*LNDAYS	0.1610***	0.5432	6.33
FLUTTER*LNDAYS	−0.0862***	0.3953	−3.38
BREATH*LNDAYS	−0.0301	0.4288	−1.34
ACHE*LNDAYS	−0.1426***	0.3615	−5.13
SWELL*LNDAYS	−0.1559***	0.3535	−7.39
PAIN*LNDAYS	−0.0182	0.4360	−0.63
NOLIM	0.9242***	1.0000	6.88
SOMELIM	0.0803***	0.4949	2.52
NOSOC	0.1243**	0.5212	3.75
ATHOME	−0.1026**	0.3854	−2.01
NEEDHELP	−0.2796***	0.2795	−7.22
INHOSP	−0.7466***	0.0000	−6.87
Utility of money function (*SQT COST)			
CONSTANT	−0.021935		−1.22
SCORE	−0.0002413**		−2.18
AGE	0.000392**		2.13
EDUCATION	0.0004650		0.46
SYMPTOMATIC	0.009791**		2.13
VIMPORTANT	−0.003691		−1.61
Scale function			
HISORE	0.1502***		2.69
HIAGE	−0.2579***		−4.09
VIMPORTANT	−0.0003		−0.01
NOPAIDLEAVE	0.1024*		1.70
HIINFORMATION	−0.0423		−0.63
FAILCHECKS	−0.2967***		−5.10
Constants			
ALPHA1	−1.2148		
ALPHA2	−0.1648		
ALPHA3	0.6591		
RHO	0.1260***		5.95
Likelihood ratio Chi-squared		536***	
McFadden <i>R</i> -squared		0.076	
Percent correctly predicted		42.4	
Number of observations		2752	

* Significant at the 10% level or better; ** significant at the 5% level or better; *** significant at the 1% level or better.

type of people who are attracted to jobs that do not offer paid sick leave over jobs that do. Finally, elderly subjects and subjects who failed more consistency checks had noisier responses, as we would expect.

DISCRETE-CHOICE ESTIMATES

As discussed above, because of potential problems and biases associated with using conditional logit choice models to estimate the discrete-choice data, we used RPL in the analysis of the choice data. Table 4 reports the RPL results. Each coefficient is estimated as a distribution defined by two parameters: a mean and a standard deviation. We obtain standard errors for both of these estimates, but report only the significance of each. Therefore, a significant mean parameter indicates that the variable is important in influencing choice, while a significant standard deviation parameter suggests that there is substantial taste heterogeneity across subjects. Although we can construct mean parameters for the omitted categories NOSE and NOLIM, there are no corresponding standard deviation estimates for those variables.

Unlike the graded-pair format, subjects evaluated the choice alternatives relative to their current health, which always was represented by the third alternative across choice sets. The subject's current health is estimated as a function of covariates using the health history data collected in the survey. Thus, in addition to the same health attribute and utility of money variables used in the graded-pair analysis, the discrete-choice RPL estimates include current health covariates and two alternative-specific constants, which indicate the probability of choosing the first or second relative to the third alternative.

We again have effects-coded the health attributes, so the coefficients on the omitted attributes NOSE and NOLIM are reported as the negative sum of the included categories. We interpret statistical significance relative to mean effects for the health attributes. Symptom variables are interacted with duration. Finally, Table 4 also reports both nominal and normalized coefficients.

Symptom was allowed to vary within pairwise comparisons in the graded-pair format but was held constant within choice sets. Because of this difference in the experimental design, we expected to see corresponding differences in estimated symptom utility weights. In particular, we expected to see less

variation among the symptom coefficients compared with the graded pairs. Instead, we observe more variation (the normalized symptom coefficients vary from 0.54 to 0.97) but lower statistical significance relative to the mean effect.ⁱⁱ Moreover, all the normalized choice coefficients are larger than the corresponding pairs coefficients. PAIN is now the worst condition and ACHE now ranked as one of the milder conditions rather than one of the more severe conditions, as indicated in the pairs estimates.

In contrast to the symptom attribute, all activity levels but AT HOME are statistically significant. Furthermore, relative to the normalized endpoints, the choice activity coefficients are larger than the corresponding pairs activity coefficients. In particular, we observe very little difference in the choice NOLIM and SOMELIM activity levels, while there is a large decline in utility between NOLIM and SOMELIM in the pairs estimates. We continue to see NOSOC rated higher than SOMELIM, but the difference again is insignificant.^{jj}

Among the symptom standard deviations, none of the standard deviations for symptoms are significant. Consequently, there is no evidence of preference heterogeneity for the symptom attributes. This result is in striking contrast to the activity standard deviations. All of these are significant and large relative to their corresponding activity categories, indicating considerable variation across observations that are not accounted for elsewhere in the model.

We observe the same pattern of signs in the marginal utility of money function as in the graded-pair model. However, the constant term VIMPOR-TANT and EDUCATION now are significant, while AGE is not. The variable VIMPOR-TANT is intended to capture subjects' attitudes toward health.^{kk} The significant negative coefficient may indicate that subjects who have strong attitudes toward health perceive that they have more personal control over their health, or that there are less expensive alternatives to treatment, leading to lower WTP. Because of differences in scaling, it is not possible to directly compare magnitudes of utility of money parameter differences between the two models.

All four of the respiratory and cardiovascular conditions in the current health function are statistically significant and indicate worse health than overall average health in the sample. ASTHMA and HEARTDISEASE were insignificant when entered independently, but significant